When Schema Lexicons Fail: Diagnosing and Repairing Entity-Layer Collapse in a Production Knowledge Base

Anonymous Author(s)

ABSTRACT

In dictionary-driven entity enrichment, the schema lexicon effectively serves as the extraction model. We report a production incident in which a category-word lexicon silently collapsed the entity layer of a 61,137-document scholarly knowledge base: 575 type-level entities ("method", "energy") carrying ∼19.2M links while every job-level monitor stayed green. We diagnose the failure across three layers (extraction mechanism, lexicon provenance, monitoring blindness), repair it with a three-source corpus-presence-attested lexicon (3,702 terms), rebuild the entity layer under production load, and derive gold-free monitors led by a concentration diagnostic invariant to uniform rescaling of link counts. A controlled experiment that fixes code and corpus and varies *only the lexicon* reproduces the collapse signature alone (109±5 entities, top-10 concentration 0.513±0.016) and isolates the repair effect (692±27, 0.300±0.008; ten resamples); the rebuilt layer carries 1,871 entities and 25.1M links; distinct concepts per document rise 15.7→37.8 with annotated-document coverage 51.5%→98.5%; a risk-stratified 300-link audit reads 0.84 overall. Matched spans are equally "valid" under old and new dictionaries (0.94 vs. 0.93)—the collapse is a vocabulary-and-mass failure invisible to per-link audits—and a composite four-leg monitor rule reaches zero misses and zero false alarms on the degradation archive. A three-domain replay on public corpora shows the failure form and the repair travel: batch-schema lexicons reproduce the collapse signature out of domain. Limitations are stated up front: the generic tail survives repair (9% of links sit on the symbol "be"), retrieval-facing gains are not attributable to the lexicon alone, annotation is model-based with single-author human verification of a 197-term stratified sample (confirm 0.934), and alarm bands are post-hoc candidate thresholds.

KEYWORDS

knowledge base construction schema quality terminology extraction entity linking production incident observability

1 INTRODUCTION

Scholarly knowledge platforms increasingly generate subject schemas with LLM assistance: a batch tool drafts entity types with a handful of aliases each, a validator checks structural well-formedness, and the schema is bound to knowledge bases as configuration [12]. When the enrichment pipeline then extracts mentions by *dictionary scan*, the schema's alias list becomes the extraction surface — the lexicon is not descriptive metadata but operative infrastructure.

The failure we report: a 61,137-document nuclear-electronics knowledge base accumulated ∼19.2M entity links onto **575 canonical entities**, every one a bare type word ("method", "parameter", "energy"); a 13,408-document physics library showed the same failure pattern at 2,018 entities. Every operational signal stayed green — jobs done, throughput nominal, link volume reading as growth. The graph had degenerated into a bipartite type–document index that was row-level indistinguishable from a populated knowledge graph.

We contribute three things. **(1) A failure mode and its controlled reproduction**: when extraction is a dictionary scan, a structurally valid schema lexicon acts as the extraction model and can silently collapse the entity layer — established forensically (§3) and then reproduced by a controlled experiment that fixes code and corpus and varies only the lexicon across ten disjoint corpus resamples, with a frozen-sample parity check tying the replica's matching semantics to the production scanner (§4). **(2) A corpus-attested lexicon lifecycle** (seed transplant, classified corpus candidates, bounded expansion — corpus presence gated) that repaired both schemas without manual lexicography and survived a full production rebuild (§5–6). **(3) Gold-free entity-quality observability**: concentration-led monitors with candidate thresholds read from 13 production libraries, reported with their known false-positive and false-negative classes (§7.1), plus a corrected incident log demonstrating that dashboard-derived incident narratives themselves require row-level verification.

2 BACKGROUND AND PRIOR ART

Dictionary- or gazetteer-driven entity recognition is one of the oldest NER traditions [6]; corpus-statistical term recognition goes back to C-value/NC-value [1] — our corpus-candidate source is that program reborn with an LLM classifier, and recent work extracts terminology directly with LLM-plus-retrieval pipelines [5]. Automatic schema and KG construction has meanwhile advanced to dynamic schema induction over corpora (AutoSchemaKG [2]), incremental schema evolution (DIAL-KG [3]), and joint evaluation of extraction quality, schema adherence, and stability [4]. Our subject is complementary and upstream of a different risk: in *dictionary-scan deployment*, where the schema's alias list is the extraction surface, we show that schema-legality and even per-link mention validity can coexist with asset-level semantic collapse — a failure locus those methods do not measure, and one that our diagnostics (distribution and name monitors, coverage accounting) target directly. We do not claim a better schema-induction or term-extraction *algorithm*; the contribution is the reproducible failure, the separation of local validity from coverage structure, and the repair lifecycle. Knowledge-graph refinement [7] and production data validation [8, 9] address adjacent problems; term-extraction evaluation is annotation-guided and domain-stratified (TermEval/ACTER [10]), and vocabulary drift is measurable even absent failure [11]. What this incident exposes is a specific failure locus: when the dictionary effectively *is* the extraction model, a structurally valid but terminologically hollow schema passed every gate *this pipeline had in place* (schema-shape validation, ingestion checks, job-level monitors). Distribution-statistics

Table 1: The collapse, measured, with snapshot provenance ("bckp." = 2026-10-06 cold backup, mid-enrichment; "disc." = discovery-day).

KB	Entities	Links
nuclear (61,137 docs)	**575** (disc.) / 503 (bckp.)	∼**19.2M** (disc.) / 4.81M (bckp.)
physics (13,408 docs)	**2,018**	2.23M (bckp.)
hydro (ref., 78 docs)	1,608	719,515

monitoring [8] is the right tradition, but the statistic must be aimed at the lexicon's terminological content; none of our gates asked whether a lexicon's entries are domain-specific terms.

Our pipeline extracts mentions in stage K1 by normalized dictionary scan (CJK substring; ASCII word-boundary with optional plural, case-folded); K2 links mentions to canonical entities by normalized form + type; K3 adjudicates typed relations with an LLM; K4 projects to the graph store. The healthy reference case in the same deployment — hydrogeology, 78 documents, a hand-evolved 3,742-term lexicon, 1,608 entities — establishes plausibility, not causation: domain, scale, and provenance all differ.

3 THE COLLAPSE, FORENSICALLY

Observation (Table 1). During an unrelated storage investigation (2026-10-08), a routine entity-quality sample returned type words where domain concepts were expected.

Mechanism. When the alias set for type *method* contains common words, every occurrence in 7.2M chunks becomes a mention, and K2 folds them onto one entity per (form, type). The mechanism executed faithfully on degenerate input; the dictionaries were the schemas — the batch generator produced a category taxonomy where a terminology was needed [13].

Timeline (compressed): 10-08 discovery → root cause, lexicon tool built/run/committed, teardown + 766,501-event re-enqueue; 10-08 night: silent mis-attribution window (below); 10-09 drain completes, statistics recomputed, two smaller libraries with the same failure pattern treated; 10-09/10: retrospective audits (scoped refresh, channel ablation, link audit, fleet-wide concentration) and manuscript. The compression is real; the evaluation is retrospective, not pre-registered.

Why nothing fired. Job-level observability was all green; link volume read as success. The observables that expose the collapse are the concentration of links on the top entities (collapsed: 0.419, measured from the pre-incident backup, versus 0.18 for the reference case) and the name distribution of those entities (pure type vocabulary). Entity cardinality per document — a factor of 2,192 below the reference case (0.0094 vs 20.6) — corroborates, but is gameable in both directions, so it is the secondary signal.

4 IS THE LEXICON THE CAUSE? ISOLATING THE VARIABLE

The repair bundled lexicon, teardown, and fleet changes; this experiment isolates the lexicon alone *before* we present the repair method: a semantic replica of the K1/K2 scan-and-link semantics with *only the dictionary swapped* — v2.0 as committed at library creation (182 scan terms), v2.5 mid-evolution (2,932), v2.1 the repair (3,733 scan terms;[1]; construction in §5), plus two v2.1 governance variants. The replica copies the scanner's matching and folding rules from the same source reading (CJK substring; ASCII word-boundary with optional plural, case-folded; fold key = normalized form + type); it is not instrumented against the live service path, so absolute levels may drift — the experiment's force is the within-frame arm contrast. A frozen-sample parity check bounds that drift: on 200 frozen KB1 chunks scanned with the *live* production dictionary (3,824 entries), every one of the 913 production mention spans is reproduced by the replica (span-level *superset*, not output equivalence: the per-term replica additionally reports 263 overlapping or nested occurrences that the production merged longest-first alternation suppresses by design, and a normalized-form bookkeeping rule differs for aliases that themselves end in "s"). Boundary, case, plural, and CJK semantics are identical; the replica's link counts are an inclusive upper bound within that frame. Because nesting prevalence can differ per lexicon, we re-ran all cross-domain arms under production-style overlap suppression (longest-match, non-overlapping): links fall 10–40% per arm but *every ordering conclusion is unchanged* (schema CS C_{10} 0.859 with identical entities; the size-matched arm C_{10} 0.323 vs. three-source 0.366; frequency still floods at 23+ concepts/doc).

Table 2: Controlled lexicon-only comparison across ten disjoint corpus resamples (mean±sd). Conc/doc = concepts per *matched* document; arms' denominators differ (977 vs. 1,579 matched docs per sample).

Arm	Links	Ents	C_{10}	Conc/doc
v2.0 collapse-era	2,969 ± 140	109 ± 5	0.513 ± .016	1.85 ± .05
v2.5 mid	4,823 ± 147	421 ± 25	0.322 ± .009	2.51 ± .05
v2.1 repair	10,042 ± 251	692 ± 27	0.300 ± .008	**3.89 ± .07**
+case-sens. short	8,758 ± 228	665 ± 28	0.265 ± .012	3.54 ± .07
+stop-word block	9,226 ± 251	691 ± 27	**0.252 ± .011**	3.67 ± .07

Design. Ten seeded, uniformly random resamples of 2,000 chunks each from the library's 5.83M visible chunks — independent draws from a *single* corpus, i.e. a sampling-stability check within one library, not cross-domain evidence — cross-sample disjoint, deduplicated on redraw (an earlier single run had sampled an ordered prefix, which we replace rather than defend). Fixed code, fixed corpus, fixed scanner; the only variable is the dictionary. The sampling unit is the chunk; in this corpus chunks map to documents almost one-to-one (sample 0: 2,000 chunks → 1,965 distinct documents, 1.02 chunks/doc), so "matched documents" below means matched chunks and no document is sampled twice. All metrics are deterministic; Table 2 reports mean±sd across the ten resamples. Concepts-per-document is averaged over *matched* documents only (≥1 scan hit under that arm's dictionary: mean 977 matched documents per sample under v2.0, 1,579 under v2.1), so the arms' denominators are different sets.

Finding 1 — the collapse signature is reproduced by the lexicon alone. Under v2.0, the same code on the same corpus

[1]Three v2.1 counts appear in this paper and refer to different objects: 3,702 canonical schema terms (the committed lexicon), 3,733 scan terms after alias dedup (the replica frame), and 3,824 entries in the production composite dictionary (schema lexicon ∪ operational seed overlay, §8's parity check).

yields 109±5 entities at concentration 0.513±0.016 with 1.85±0.05 concepts per *matched* document: the collapse needs no pipeline bug, no data loss, no misconfiguration — only a category-word dictionary.

Finding 2 — the repair effect is isolated and monotone. v2.1 multiplies link-bearing entities ×6.3 and concepts per matched document ×2.1 (matched docs per sample 977→1,579); generic-head mass falls 0.146→0.058. The production collapsed concentration (C_{10}=0.419) sits between the v2.0 (0.513) and v2.5 (0.322) counterfactuals, consistent with mid-enrichment dictionary evolution. These are *structural* signatures: entity count, concentration, and concepts per document do not by themselves measure terminology recall or sense fit — the only semantic reading in this paper is the link audit of §7.

Finding 3 — the be class is treatable at known cost. Case-sensitive short symbols remove the largest entity at the price of match volume (links −12.8%; C_{10} →0.265); a stop-word block is gentler (links −8.1%, C_{10} 0.252). The deltas measure matching *output*, not true recall: the removed mass mixes errors with correct terms, and the 0.580 pre-intervention judged-valid rate on this class (§7) motivates the treatment — the removed-vs-retained verification below prices exactly this cut. *Provenance*: decomposing v2.1's link mass by which committed lexicon first contained each term — v2.0-inherited 22.0%, mid-evolution 16.0%, **three-source-new 61.9%** (384 of 686 entities) — is a decomposition, not a per-source causal ablation; the public-domain cached-component ablation (§8) supplies the single-source analogue, while independently regenerating single-source lexicons for the *production* library remains future work. And the arms vary the dictionary as a whole (~20× size gap), so term count, selection quality, and per-source method are not separately identified — the equal-size frequency control and the blind old/new matched-context judging below are the two controls that close this gap on the same frozen samples.

Budget sweep, closing the size confound. A natural concern is whether the isolated comparison (182-term collapsed vs. 3,733-term repair dictionary) confounds size with selection. We re-ran the frozen KB1 frame at fixed budgets $K \in \{200, 500, 1{,}000, 2{,}000, 3{,}733\}$ with three same-K sources: the v2.1 pipeline in construction order (prefix), frequency-top-K, and C-value-top-K (candidates ranked on sample-0 text; frame overlap disclosed). Coverage under the dual-annotated reference grows monotonically with v2.1's own budget (R 0.004→0.043 from K=200 to 3,733) while concentration falls 0.73→0.30 — the repair's value accrues with budget but its structure at every budget is distinct: frequency-top recruits 4–5× the entities (2,036 at K=2,000) for nominally higher R that rides on mass its own blind validity disqualifies (0.45 at the equal-size control), and C-value-top saturates at 554 entities with R stuck at 0.012 — statistical salience without attestation stalls on this corpus. Re-running the sweep with the baselines re-typed by the unified classifier leaves R identical at every budget (entity counts within 1.5%)—typing- invariant. Size explains the counts; it does not explain the validity or the distribution.

Verification: semantics, governance, monitors. Three follow-ups on the same frozen samples close the declared gaps. (i) *Blind term-mention judging* (entity-uniform; judge Qwen3.6-35B, temp 0, disclosed family): old and new dictionaries are *locally* indistinguishable (0.940 [0.890, 0.968] vs. 0.927 [0.873, 0.959] of matched spans are genuine term mentions) — the collapse is not per-link falsehood but a vocabulary-and-mass failure, which is why per-link auditing alone could not have caught it; an *equal-size frequency-only control* (3,733 terms, top corpus frequency, no selection) reads 0.450 [0.356, 0.548] — size does not buy validity; selection does. (ii) *Governance verified*: short-symbol hits removed by case-sensitivity are 0.033 [0.009, 0.114] valid vs. 0.700 [0.575, 0.801] for retained ones — the −12.8% link cut removes almost purely error mass (retained still carries 30%). (iii) *Graded degradation* (replacement axis: drop m% of v2.1, keep all category words): the C_{10} band fires at ≥75% substitution (0.371) and the full collapse reproduces (0.513, bitwise-identical arm); random shrinkage fires count monitors first (links −50% at 50% shrinkage) while C_{10} stays flat to 50%; moderate substitution (10–50%) is a blind zone for both — no single gold-free monitor covers it; the stack deploys as a set.

Short-symbol ambiguity: four linking policies. On 150 short-symbol mentions sampled from the frozen replay chunks (60 on the be class; each mention judged once, blind, by judge A and the cross-family judge, 12 disagreements), four policies on the same dictionary: linking everything (current behavior) reads precision **0.24**; exact-case gating (the deployed treatment) 0.725 at true-symbol recall 0.806, rejecting 73% (rejection precision 0.936); adding context evidence (long-term neighbors) recalls slightly more (0.833) at lower precision (0.652); context plus reverse-NIL trades recall (0.583) for precision (0.724). On the be class specifically, *zero* of 60 sampled mentions are genuine element references — the exact-case gate removes pure error there. Context gating and case gating exchange precision for recall rather than dominating each other; the recovered mass lives in the non-be short symbols.

Why 692 entities here vs 1,871 in production (§7)? The isolated frame counts only entities induced by scanning 2,000 chunks (0.034% of the library) under replica folding: rare terms need corpus mass to be attested, and the production layer additionally carries entities from the full corpus and historical merging. The causal claim rests on within-frame comparisons, never on this cross-frame count.

5 CORPUS-ATTESTED LEXICON CONSTRUCTION

The repair lexicon whose causal role §4 isolated is built from three sources, all gated by corpus presence (a term the corpus never attests is not admitted — presence is necessary, not sufficient, for termhood): **seed transplant** (2,284 terms from a healthy near-domain library); **corpus-candidate classification** (top 8,000 unigram/bigram n-grams by chunk frequency over a 3,000-chunk-per-KB sample, frequency ≥5 after boilerplate and stop-word filtering; an LLM classifies each into a schema type or rejects it); **per-type expansion** (≤60 per type). Catch-all types accept no classified noise; cross-type conflicts resolve by specificity. Classifier: Qwen3.6-35B-A3B (AWQ 4-bit, community build; served under a historical alias), temperature 0.2, single

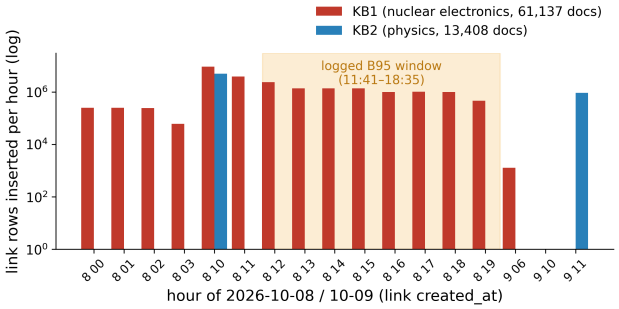


Figure 1: Rebuild telemetry from the link registry's own timestamps. The shaded band is the *logged* silent window; inserts continue through it — the timestamp correction of §6.

idempotent run. Output: nuclear v2.1 = 3,702 canonical terms; physics v1.1 = 1,279.

6 REBUILD AT SCALE — AND A CORRECTED INCIDENT LOG

Per knowledge base: delete graph vertices with edges, the four relational tables in bounded chunks, and the search-side projection; re-enqueue all documents through four stages. The printed "766,501 jobs" counts *events*, not distinct doc-stages (74,545×4 = 298,180): the surplus is the repair window's re-enqueue waves (silent-empty reset 75,831; zero-effect relations/graph resets ≈149k; watchdog-deletion recovery ≈122k; small libraries 5,772).

Four pipeline pathologies surfaced and were fixed with measured effect — an unbounded projection join that saturated the database (25+ min → seconds once scoped), two link-insertion deadlock classes (deadlocks 1213 → 0), and an $O(N^2)$ alias check the richer lexicon itself amplified (fleet link throughput 8/min → 1,100/min, a window-level aggregate). The archival version carries the full per-pathology accounting.

The fifth pathology, revisited. Our log recorded a parameter-order bug in a single-statement link writer whose rows `INSERT IGNORE` "silently swallowed" for seven hours. A minimal reproduction on the production engine (MySQL 8.4, `STRICT_TRANS_TABLES`, faithful DDL) sharpens the mechanism: the NULL from `UNHEX('alloy')` is **replaced by the implicit default 0x00×16 and the row is inserted** — into a ghost bucket invisible to every per-KB query — and a later correct re-insert for the same (mention, entity) pair is **silently skipped by the unique key**: the orphan poisons the repair unless deleted first (production carries zero 0x00 rows today, so cleanup happened). The duration claim also fails row-level forensics: the registry's own `created_at` histogram (Figure 1) shows KB1 inserting at 1.0–2.3M rows/hour *through* the nominal silent window and KB2's entire link layer landing in one hour. We keep the log entry and its correction side by side: incident narratives written from job dashboards are themselves unverified claims.

Table 3: Entity layer before/after, with snapshot provenance.

Metric (KB1)	Collapsed	Repaired
canonical entities	575 / 503 (10-06)	**1,871**
entity links	~19.2M / 4.81M	**25.1M**
typed relations	~0	**44,043**
top-10 concentration	0.419	**0.345**
concepts/doc (all-doc)	15.7 (51.5% covered)	**37.8** (98.5% covered)
KB2: concepts/doc	29.5	**61.4**

7 EVALUATION

Entity layer, same-snapshot framing. The before/after rows mix two pre-repair snapshots; the like-for-like comparison is the 10-06 cold backup vs. post-repair (503→1,871 entities, 4.81M→25.1M links); the discovery-day figures (575, ~19.2M) are reported for the incident record, as mid-enrichment evolution continued after the backup. Concepts-per-document replaces the raw entity/document ratio: its numerator scales with documents and the coverage column exposes the collapse's most user-visible symptom — half the library had *no* annotations. Absolute link counts *rise* after repair (KB1 19.2M→25.1M; KB2 2.23M→5.0M): real terms match more often. Count was never the quality signal; distribution and names are.

Name quality (Figure 2). The repaired head is still generic: `be` (2.26M links, 9.0%), `alpha`, `gamma`, `beta`, `method`. The `be` case is a normalization failure — element symbol and auxiliary verb merge under case-folded matching — and §4 prices its treatment. The repair moved the problem from "no terminology at all" to "terminology plus a generic tail."

Link audit: 300 risk-stratified links. Entities drawn uniformly within four strata (top-10 by frequency; medium, ranks 11–100; tail, ranks 101+; short symbols, canonical names of ≤4 letters), then one link uniformly per entity — seeded and replayable. Span integrity 300/300. The judged-valid rate is **0.837** for this sample (Wilson 95% CI [0.791, 0.874]; entity-cluster bootstrap over the 187 clusters, 10,000 resamples: [0.782, 0.887]); by stratum: top-10 **0.920** [0.812, 0.968], medium 0.850 [0.767, 0.907], tail 0.910 [0.838, 0.952], short-symbol **0.580** [0.442, 0.706]. Three scope notes. First, the 50/100/100/50 quotas target risk, not production link frequencies: 0.837 is a stratum-mixed rate and must not be read as an unweighted production-wide link precision. Second, the 300 links come from 187 distinct entities (the head stratum's 50 links from just six; the short and head strata share "term"), so the intervals treat links, not concepts, as exchangeable — the CI is neither an interval for the 25.1M production links nor for 187 independent concepts. Third, the design's purpose is to localize high-risk failure — which it does: the short-ambiguous-symbol class that Figure 2 colors red and §4 prices is where validity collapses — a more informative finding than any overall figure. A second Qwen-series model re-adjudicated the earlier uniform sample: on the 55-link parseable subset agreement is 96.4% (domain 14/14) — a cross-model check, not family-independent confirmation (Figure 3), and unparseable responses may be non-random. Human adjudication remains the decisive future control.

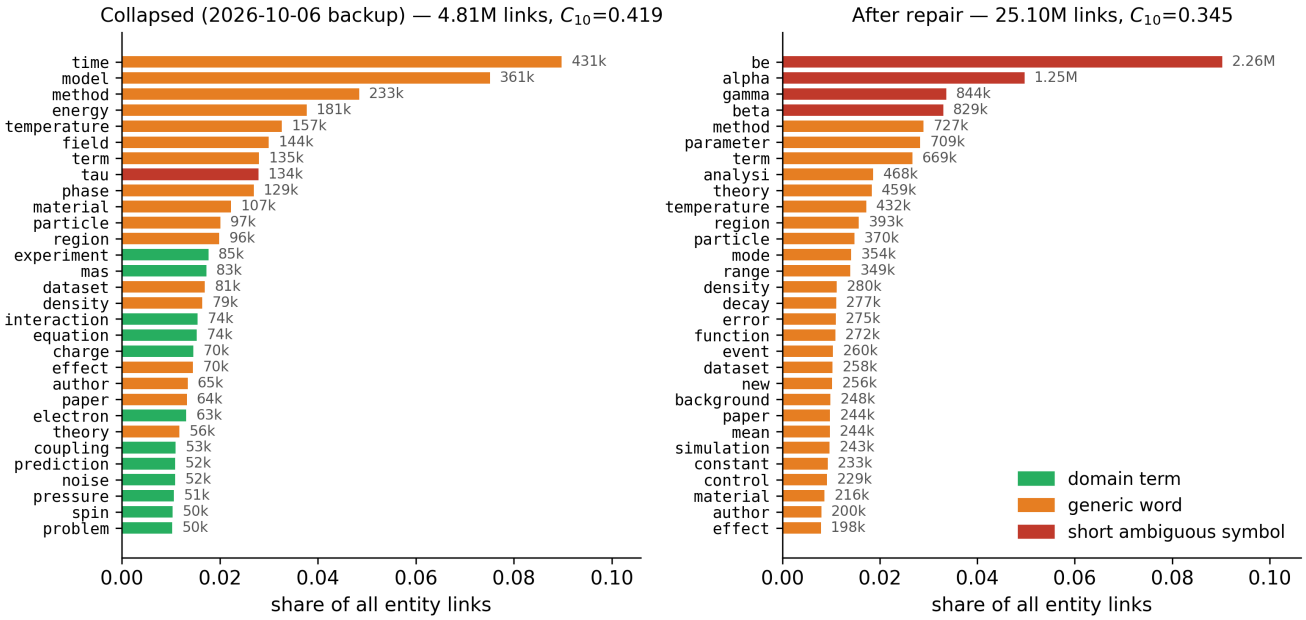


Figure 2: Top-30 entities by link share, before (cold backup; C_{10}=0.419) and after repair (C_{10}=0.345). Categories follow a fixed three-way rubric (domain term / generic academic word / short ambiguous symbol), assigned by the authors per entity; the rubric and per-entity assignments ship with the artifact.

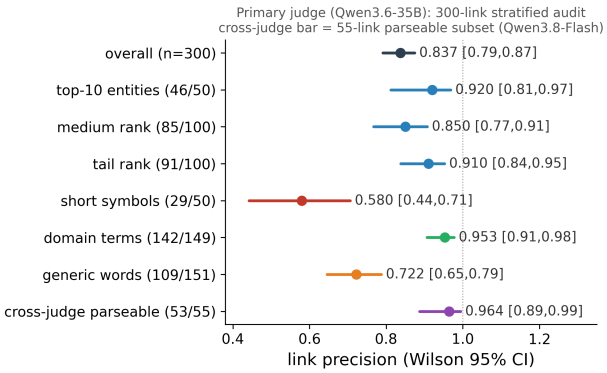


Figure 3: Judged-valid rates with Wilson 95% CIs: the risk-stratified 300-link sample (stratum-mixed, 187 entities; primary judge); the cross-judge bar applies to the 55-link parseable subset of the earlier uniform sample.

Table 4: The candidate calibration behind Figure 4, summarized (the collapsed row is the pre-repair backup, not a live point).

Group	Libs	Entities	C_{10} range
Collapsed (backup)	1	503	0.419
Hand-curated control	1	1,239	0.184
Other prod., ≥500 ents	4	874–2,482	0.259–0.287
Rebuilt here, large	2	1,871 / 2,726	0.162–0.345
Rebuilt here, small	2	153–629	0.398–0.768
Other prod., <500 ents	4	5–424	0.395–1.000

Snapshot 2026-10-12; the rebuild-completion count for the 2,726-entity library is 2,730 (four-entity post-rebuild drift), cf. §6.

Retrieval: honest attribution. Scoped coverage rose +6.6/+7.3 points over the repair window, but three controls bound the attribution: (a) removing the direct entity channels under the final configuration leaves @10 any-hit and document coverage unchanged on both scoped sets (MRR +0.0006, CI [0, 0.0018]; cov@20 −1.0pt on one set) while cutting p50 latency 1.46 s→0.52 s — and a forced-trigger check on the natural-phrasing set shows the channels are alive (54/99 fired; killing them costs 20 losses / 0 wins), so the ablation bounds these channels' marginal contribution in this configuration, not the semantic assets' value at large; (b) alias expansion, the one entity-adjacent mechanism, is +1.5pt (ns) on the benchmarks at 2× latency; (c) two same-configuration runs differ on exactly one query (0.5pt noise floor). The trajectory is system-level; the entity layer's retrieval-facing value runs through asset surfaces (documented in the companion system manuscript [14]), not fusion votes.

7.1 Monitors and Discussion

Four gold-free monitors, in the spirit of production data validation [8]: **(a) top-10 link concentration** (primary): invariant to uniform rescaling of counts, but bounded below by $\min(10, N)/N$

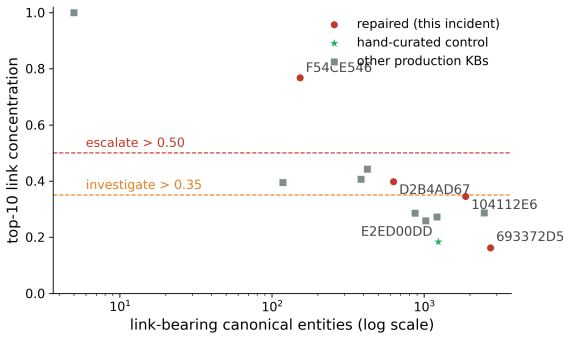


Figure 4: Top-10 link concentration vs. link-bearing entities across 13 production libraries; red = rebuilt in this incident, green star = hand-curated reference. Bands are candidate thresholds read post-hoc from this deployment.

— trivially 1 for a library with ≤10 entities — and sensitive to topic concentration as well as to degradation. Across the 13 deployment libraries the metric spans 0.16–0.77 among libraries with ≥100 entities (the smallest, five entities, reads 1.000 — the floor bound, not degradation; Table 4); healthy/repaired large libraries run 0.16–0.19, the rebuilt nuclear library 0.345, the two small treated libraries *currently* 0.398 and 0.768 (reported as measured). The collapsed physics backup reads 0.290 — *below* the 0.35 band — so a concentration threshold alone does not catch every collapse; the name-audit and coverage legs carry the rest. The bands (>0.35 investigate, >0.5 escalate) are *empirical, post-hoc* candidate thresholds read from these 13 libraries — triage triggers for human inspection, not classifiers, and not out-of-sample validated. Their known blind spots, stated as such: *false positives* — healthy libraries whose terminology is genuinely concentrated, e.g. the two small treated libraries at 0.398/0.768 (real terms, small corpora) and the public-domain CS three-source arm at 0.404 (§8); *false negatives* — replacement-type degradation below ~75% substitution, where C_{10} moves less than one sd (graded degradation), and any collapse whose head happens to be diversified. Validation need not re-collapse production: the graded-degradation arm above injects lexicon decay offline into a frozen slice and reports fire bands and size sensitivity (the experiment has no time axis, so we make no detection-latency claim). A leave-one-library-out check on the 14 rows (thresholds re-frozen at the q90/q97 of the remaining healthy libraries each fold) yields 12 correct quiets, the same known false-positive class (the genuinely concentrated small library), and the same known false-negative class (the collapsed backup reads below the LOO band — count monitors and the name audit carry it): the bands are self-consistent across libraries but remain deployment-derived; unseen-deployment validation stays future work. *Composite rule, evaluated.* Because single-leg thresholds misfire in both directions, the deployable unit is a four-leg disjunction (fire on any leg): concentration (C_{10}>0.35, small libraries exempt), link-count drop (>50% vs. healthy baseline), generic-head share (>0.10), and entity-cardinality drop (>50%). On the frozen graded-degradation archive this composite fires on every should-alarm state (collapse, ≥75% replacement, ≥50% shrinkage) with *zero* misses and zero false alarms on the healthy baseline and mild degradations — reported as in-sample diagnostic verification (thresholds and classes derive from the same libraries; frozen-threshold unseen-library validation is open); the 10–50% replacement band remains undetected by every leg (at 50% the generic-head leg sits at its threshold, 0.096 vs. 0.10) — the honest structural floor of aggregate-statistics monitoring, and the reason the name-audit leg exists. **(b) entity cardinality ratio** (coarse secondary, gameable both ways). **(c) top-entity name audit** (five-minute quarterly control). **(d) target-table growth reconciliation** — the B95 ghost-bucket mechanism makes this mandatory wherever a bulk writer is guarded by `INSERT IGNORE`. The small-library treatment is an illustrative within-deployment case (the candidate bands were read after observing these 13 libraries — not out-of-sample validation): cardinality normalized (870→629, 209→153) while concentration still flags both — the two monitors separate "cardinality repaired" from "distribution still skewed."

The incident is not an argument for LLM-everywhere: if extraction is a scan, the dictionary serves as the model and deserves a trained model's evaluation discipline. Admission criteria (corpus presence) are necessary but not sufficient; *ranking* criteria (IDF-style contrast, bilingual consistency, length/type priors) are the open problem — the generic tail is the standing evidence.

8 DOES IT GENERALIZE? THREE PUBLIC DOMAINS

The incident corpus is nuclear-science literature; a natural objection is that the failure and the repair are local accidents. We therefore replay the lexicon-only design on three public domains — arXiv cs.CL (10,000 abstracts), PubMed biomedicine (9,988), and arXiv cond-mat.mtrl-sci (10,000) — with seeded splits per domain: 3,000 abstracts for lexicon construction only, 100 frozen test abstracts for silver annotation, and the remainder replayed as ten disjoint ~690-document samples for structural metrics (mean across samples; sd archived). *Mass* throughout denotes two reported quantities — lexicon size (|L|) and per-document annotation volume (concepts/doc) — plus matched-document coverage where stated; we report all three per arm and predefine no scalar utility over them, so no "best" is asserted. Five arms share one scanner; only the lexicon varies: *batch schema* (the production-failure form: each domain's *own* current repository schema lexicon, i.e. the batch tool's committed output for that domain — category-shaped alias sets, in CS largely Chinese or uppercase-acronym forms), *frequency* top-K and *C-value* top-K (each size-matched to ours), *LLM direct* (per-batch extraction without corpus validation), and the *three-source* recipe. One adaptation is disclosed: source 1 transplants seeds from a healthy near-domain library, which does not exist for these public domains — we substitute LLM knowledge seeds under the same corpus-presence validation; sources 2–3 (corpus-candidate classification, per-type expansion with presence validation) are unchanged.

Model-annotation protocol with human verification. The annotation layer is model-based; human input enters as verification of a stratified sample (below), not as from-scratch labels. Silver reference lists come from a strong annotator LLM

(judge A, Qwen3.6-35B) enumerating every domain term in each test abstract; arm quality is then judged blind on stratified samples (head/medium/ tail/short-symbol entities) by judge A and a second Qwen-series model (judge B), with disagreements and missing judgments arbitrated by a third *cross-family* judge (DeepSeek; 8–29% of rows per cell, Wilson intervals below). Two instrument limits are disclosed rather than hidden. First, enumeration is the weak link: on 30-document subsets re-annotated by the cross-family model, term-list Jaccard with judge A is 0.41 (CS), 0.55 (biomedicine), 0.52 (materials) — so silver P/R are indicative levels, not ground truth, and we read contrasts between arms (which share the instrument) rather than absolute magnitudes. Second, judge B cannot enumerate at all (the gateway forces chain-of-thought past read timeouts) and is used only for short blind judgments; and the LLM-direct arm's silver-P is same-family inflated, since judge A's family also built that arm.

Instrument sensitivity, tested on the shared subset. Because a shared instrument does not guarantee arm-wise bias cancellation, we recomputed all five arms' occurrence-level P/R on the exact 30-document A∩C subset under *both* references (matching rule: lower-cased surface with naive plural stripping, applied symmetrically). The recall ordering of the five arms is **identical under both judges in all three domains**; the F1 ordering is identical in biomedicine and materials, while in CS only the middle band permutes (frequency/C-value/three-source swap ranks 2–4; first and last are unchanged). Three-source's precision *rises* under the cross-family reference in all domains (CS 0.218→0.312, biomedicine 0.195→0.287, materials 0.246→0.260) while LLM-direct's falls (CS 0.653→0.544), consistent with the disclosed family bias and mildly favoring three-source under the harder reference. We therefore rest cross-domain conclusions on (i) structural metrics (no instrument), (ii) recall ordering and (iii) arbitrated blind validity, and treat mid-band F1 orderings and absolute P/R levels as instrument-conditional. Judged-valid rates carry Wilson 95% CIs — e.g. biomedicine batch-schema 0.906 [0.810, 0.956] vs. three-source 0.850 [0.756, 0.912]: overlapping — and the arbitration fraction per cell is 0.08–0.29. *A dual-annotated reference, and what survives it.* To strengthen the instrument we finally cross-annotated 200 complete units — 80 frozen KB1 chunks plus 40 test documents per public domain — with two independent model annotators reading full text (not approving candidates): span-level entries with termness, specificity, type, and identity-ambiguity flags. P1 = deepseek-flash, P2 = deepseek-v4-pro (same vendor, different models — overlap disclosed; a gateway thinking-mode model was attempted first and is unusable for full-document annotation: forced deliberation exceeds any practical token budget), disagreements arbitrated by the cross-family Qwen judge. Agreement is materially stronger than single-model enumeration: domain-set Jaccard **0.59–0.66** across the four scopes, termness agreement on overlap **94.5–98.8%** (vs. 0.41–0.55 for enumeration pairs). Under this reference: the batch-schema arms recall 0–1% (the collapse signature under the strongest instrument yet); the production lexicons' first recall estimate on KB1 reads v2.0 1.0% → v2.1 4.3% of adjudicated distinct concepts (4.3×; the reference is deliberately exhaustive, so absolute levels are floors, not ceilings); the selection arms' recall deficit replicates (learned/cover 0.7–3.2%); and the frequency arm's apparent recall rides on mass its own blind validity (0.29–0.40) already disqualifies. *Human verification*: one author independently confirmed a seeded, stratified sample of 197 adjudicated terms over 20 units (5 per scope; sheet, protocol, and per-row verdicts in the artifact) — confirm rate **0.934** [0.890, 0.961]; by scope 0.833 (KB1) / 0.967 / 0.925 / 0.950; two-model consensus items verify at 0.949 vs. 0.853–0.960 for arbitrated additions. This bounds the reference's precision from the model side; it does not measure omissions (verification, not from-scratch re-annotation) and is single-author — an independent two-annotator gold remains future work. The protocol, adjudications, and per-unit records ship with the artifact. Entity-cluster bootstrap CIs for the three-source validity rates (10,000 resamples) are [0.747, 0.921] (CS), [0.772, 0.924] (biomedicine), [0.846, 0.974] (materials) — wider than link-level Wilson intervals, conclusions unchanged.

From-scratch enumeration (recall side): the same author enumerated terms from zero on 30 further held-out abstracts (10 per domain, frozen test splits; 418 unique terms, exact case-folded surface match). The batch-schema arm covers **0/418** — 0 even among the 83 terms recurring in >10 construction-corpus documents; size-aligned statistical arms cover 16.9–22.9% of that core stratum (C-value 19 > frequency = three-source 14), LLM-direct 47.0% with a 4.6× larger unvalidated lexicon. Half the terms (209) never occur in the 3,000-doc construction corpus — paper coinages that no bounded static lexicon can carry (mention-layer scope), bounding lexicon-layer recall itself. Per 1,000 entries, recall efficiency: C-value 16.6 > three-source 14.8 > frequency 12.2 > LLM-direct 10.5; the strata table ships with the artifact.

The collapse signature travels (Table 5, Figure 5). The batch-schema arm on CS is the incident in miniature: the domain's *own* committed schema lexicon yields C_{10}=0.899 with 18 matched entities and 5.7% of documents carrying any annotation (its aliases are category-shaped and largely Chinese or uppercase-acronym forms, absent from English abstracts); silver recall is 0.000; a language-aligned control (below) rules out orthographic confounds. Biomedicine and materials schema arms are milder — some of their category words ("disease", "cell", "phase") happen to be usable domain terms there — but still concentrate (0.43–0.83) at recall ≤0.009. Frequency floods at any size: ∼30 concepts per document with validity 0.29–0.40 — volume without terminology. C-value at equal size repairs concentration (0.24–0.46) but not validity (0.27–0.37): in abstract-length corpora, statistical salience is not termhood. Since these baselines' types were randomly assigned, we re-ran both under *unified typing* (term sets frozen; types from the same LLM classifier as three-source's classification stage). Unified typing shifts their validity by −3 to +13 points (frequency 0.405/0.362/0.450, C-value 0.400/0.400/0.333 in the three domains, vs. 0.267–0.400 random)—a bounded confound that nowhere closes the gap to three-source (0.838/0.850/0.912). The dominant deficit is candidate selection: the same classifier accepts only 6–19% of the frequency arm's terms. LLM-direct reaches the highest validity in materials (0.925 [0.846, 0.965]) and the highest silver-P everywhere — but at 4.7–8× the lexicon size with no presence control, and its silver-P is the same-family-inflated column. Among the three *size-matched* arms, three-source holds the best arbitrated validity in CS (0.838 [0.742, 0.903]) and biomedicine (0.850 [0.756, 0.912]) and sits within overlapping intervals of

Table 5: Cross-domain replay: five lexicon arms on identical frozen samples (structural metrics are means over ten disjoint ~690-document samples; |lex| size-matched for freq/cvalue/threesource). Valid = blind judged-valid rate after cross-family arbitration (n=60–80 per cell; Wilson 95% CIs in the text); P/R/F1 are occurrence-level against silver enumerations — levels indicative only, with arm recall orderings judge-stable (see text). Hatched arm in Figure 5: silver-P same-family inflated.

Domain	Arm	Terms	Entities	C_{10}	Conc/doc	Valid	P_s	R_s	F_1
CS	batch schema	95	18.4	**0.899**	1.14	0.565	0.000	0.000	0.000
CS	frequency K=251	251	312.4	0.227	30.7	0.293	0.101	0.097	0.099
CS	C-value K=251	251	273.4	0.457	7.2	0.267	0.300	0.043	0.075
CS	LLM direct	1198	344.5	0.501	4.6	0.787	0.663	0.132	0.220
CS	three-source	251	202.1	0.404	3.6	**0.838**	0.274	0.050	0.085
Biomed	batch schema	130	100.5	0.428	2.8	0.906	0.089	0.009	0.016
Biomed	frequency K=513	513	642.7	0.114	29.7	0.362	0.038	0.039	0.038
Biomed	C-value K=513	513	495.4	0.249	8.1	0.350	0.073	0.022	0.034
Biomed	LLM direct	2003	591.5	0.309	4.9	0.838	0.343	0.085	0.136
Biomed	three-source	513	408.7	0.281	5.1	**0.850**	0.273	0.060	0.098
Materials	batch schema	152	58.5	**0.829**	2.3	0.688	0.054	0.004	0.007
Materials	frequency K=384	384	481.4	0.131	31.8	0.400	0.029	0.035	0.032
Materials	C-value K=384	384	411.2	0.238	7.5	0.367	0.155	0.044	0.069
Materials	LLM direct	2045	713.8	0.198	6.9	**0.925**	0.336	0.093	0.146
Materials	three-source	384	337.4	0.268	5.4	0.912	0.223	0.046	0.076

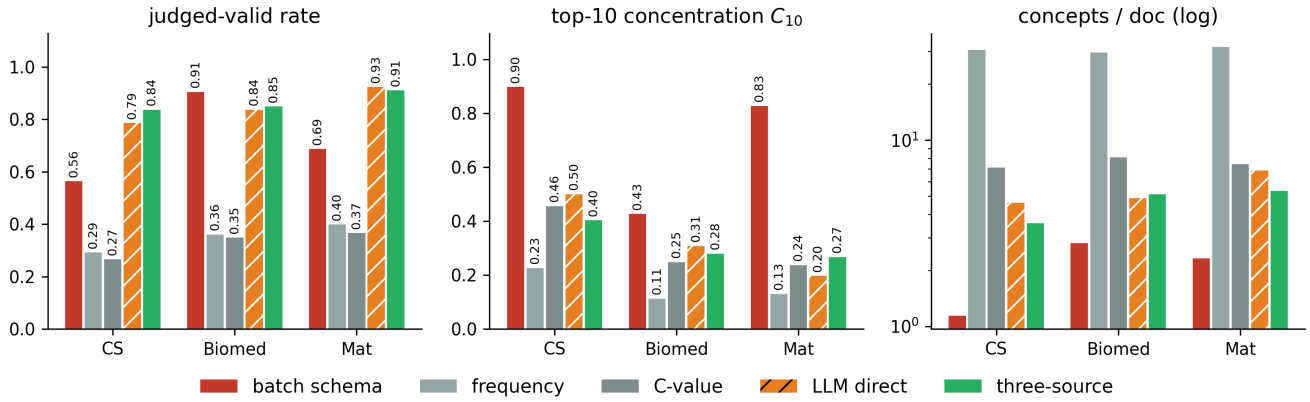


Figure 5: Cross-domain replay at a glance. Left: blind judged-valid rate after cross-family arbitration. Middle: top-10 concentration (the collapse signature; note the schema arm). Right: concepts per document on a log scale (the flooding signature; note the frequency arm). The LLM-direct arm is hatched where its silver-P reads same-family inflated.

LLM-direct in materials (0.912 [0.830, 0.957] vs. 0.925) and of batch-schema in biomedicine (0.850 vs. 0.906 [0.810, 0.956] — whose recall is 0.009, so its validity lead does not qualify it as a lexicon); its recall is at or above the frequency baseline in two of three domains at one-half the C-value arm's and one-eighth the frequency arm's annotation volume per document. Generic-head share is 0.5% (CS), 3.4% (biomedicine), 7.0% (materials) — reported per domain because the head list is the incident corpus's category vocabulary: its biomedicine/materials drivers ("disease", "drug", "phase", "cell") are legitimate domain terms in those domains, so the cross-domain residual is substantially list–domain mismatch rather than pure genericity (the frequency arm reads 4.5% in CS, where the same words are generic). No single arm dominates all reported metrics; we present three-source as the design point that pairs size-matched validity with bounded annotation volume under this protocol, not as a uniquely optimal balance.

Language confound, resolved by control. Because the CS batch-schema lexicon carries Chinese and uppercase-acronym aliases against English abstracts, its failure could be read as language mismatch rather than semantic shape. A language-aligned control (English-normalized aliases only: CJK dropped, ASCII lowercased, 80 terms) reads C_{10} 0.859 with the same 18.4 entities and the same 5.7% matched documents as the raw arm — the Chinese aliases never matched English text, and the English ones that do are category words. Normalization does move one observable: raw 94.1 links fall to 65.9 (−30%), because case-variant aliases compile to overlapping case-insensitive patterns that each counted the same span; folding duplicates deflates top-head counts unevenly

— the whole of the C_{10} shift. A translation control closes the remaining path: the schema's non-ASCII aliases are exactly sixteen type-name translations (the Chinese for "algorithm", "data structure", etc.), not domain-term aliases; translating them to English adds two net-new surfaces with zero corpus hits—every metric identical to the deletion-based alignment. There is no Chinese terminology to lose: the CS collapse is semantic, not orthographic.

What carries the recipe (CS ablation, same frozen samples; Table 6). Removing corpus-candidate classification collapses coverage (concepts/doc 3.60→1.63): classification is the coverage workhorse. Removing presence validation keeps 977 raw entries for almost the same coverage (3.92 vs 3.60) — the gate removes 74% of raw LLM mass at ~8% coverage cost, which is precisely the trade the incident demanded. Seed and expansion alone are thin (1.3–1.5 concepts/doc); unvalidated knowledge seeds carry 13× the entries of validated ones (768 vs 58) for barely more matched entities (80.8 vs 50.9) — mass without coverage. The size-matched frequency control floods (30.7 concepts/doc, 4.5% generic head). The recipe's value is in the assembly and the gate, not in any single source. One component is demoted on this evidence: *expansion*. Removing it costs nothing structurally (entities 211.7 vs. 202.1, concepts/doc 3.73 vs. 3.60), and its standalone hits read 0.433 valid under double judging with only 0.50 judge agreement (n=60) — contested quality. We reframe expansion as an optional diversity module, not a necessary source; the load-bearing pair is corpus-attested classification plus the presence gate.

8.1 Where Recall Is Lost, and What Selection Can Buy

The silver recalls above are occurrence-level against an enumeration instrument; before optimizing selection we decompose where the loss sits. For every silver occurrence we test four nested events: expressible by *some* candidate (C: a wide pool — all construct n-grams at frequency ≥2 plus definition-evidenced abbreviations plus all LLM candidate lists; 12,149/15,032/13,121 terms per domain), kept in the arm's dictionary (K), detected by the actual scan at that occurrence (D), and linked under normalization (L). The funnel is unambiguous: $P(C)$ is 0.44/0.45/0.41 (CS/biomed/materials) — *the candidate pool, not the ranking, is the binding constraint*; three-source keeps $P(K|C)$=0.15/0.21/0.14; scan and link are nearly lossless ($P(D|K{\cdot}C)$ 0.89–0.98, $P(L)$ ≈1). The pool gaps are two-character element symbols (excluded by the tokenizer's 3-character minimum), terms attested only in test abstracts (construct-side presence cannot see them), and neologisms; the selection gaps include, e.g., *LLM* and *large language model* themselves in CS.

Risk-constrained selection, tested and not adopted. The natural next question is whether a learned or set-selection objective beats per-term ranking at equal budget. On the same pools, with construct-side labels from the cross-family judge (test documents never touch pool, training, or tuning) and proxy link-correctness scores (explicitly proxies), neither learned ranking nor coverage-linked selection dominates the three-source pipeline at any tested budget ($K/2$ to $2K$): silver-P rises while arbitrated blind validity collapses in CS and biomedicine (learned-rank 0.29 [0.19, 0.41] and 0.46 [0.32, 0.55] vs. 0.838/0.850) and trails in materials (0.86–0.89 vs. 0.912 at one-fifth the recall); recall is flat in K (the pool ceiling again); and the selected lexicons *concentrate* (C_{10} 0.69–0.73 in CS vs. 0.404) — a proxy-validity objective re-creates the head-heavy distribution our monitors flag, and the silver-P gain was an instrument artifact (enumeration under-cover flatters arms whose extra mass is invalid). We keep the three-source pipeline as the repair baseline and report this as a bounded negative result; construction details, budget curves, and the beam-search reference (within 1.5% of local search) are in the archival long version and the artifact.

Table 6: Source ablation on CS (identical frozen samples; terms column makes the size confound explicit). −presence = all raw components unvalidated.

Arm	Terms	Entities	C_{10}	Conc/doc	Gen-head
full three-source	251	202.1	0.404	3.60	0.005
−seed	221	194.1	0.437	3.35	0.005
−classify	93	70.2	0.601	1.63	0.000
−expand	232	211.7	0.397	3.73	0.004
−presence (raw)	977	260.9	0.374	3.92	0.004
seed-only (valid.)	58	50.9	0.666	1.52	0.000
seed-only (raw)	768	80.8	0.616	1.57	0.000
classify-only	181	169.0	0.463	3.18	0.005
expand-only	42	27.3	0.838	1.29	0.000
freq size-matched	251	312.4	0.227	30.7	0.045

9 CONCLUSION

When does a structurally valid lexicon produce semantic collapse? When its entries are category words: the scan executes faithfully, and a dictionary-only swap reproduces the signature. **What does corpus-presence gating repair, and what remains?** Terminology and coverage (annotated documents ~51.5%→98.5%) — but a generic tail persists; presence is necessary, not sufficient. **What can gold-free diagnostics see?** Distribution and names; per-link truth they cannot see — blind judging shows old-dictionary links are locally as "valid" as new ones (0.94 vs 0.93), so the collapse hides from per-link audits, and the semantic gain runs through vocabulary and mass (equal-size frequency control: 0.45 vs 0.93). The incident motivated and validated; the controlled swap, the stratified audit, the graded-degradation alarms, and a three-domain public-corpus replay carry the evidence. **Does it generalize?** On three public domains the batch-schema form reproduces the collapse signature (C_{10} up to 0.899 with recall ≤0.009), frequency floods, and the three-source recipe is the size-matched design point with the best arbitrated validity at bounded annotation volume — a trade-off under the tested metrics and AI instrument, not a claimed unique optimum; the failure mode and the repair are properties of the construction method, not of one corpus.

10 LIMITATIONS

(1) The generic tail survives repair (9% of links on "be"). (2) Retrieval attribution is bounded, not eliminated: the full lexicon×configuration decomposition is runnable on frozen slices or isolated stacks via graded degradation — not yet run, rather than impossible. (3) Coverage: two libraries rebuilt, two smaller treated; bilingual anomalies documented, not explained. (4) Six

secondary incidents extended the repair; reported as findings. (5) Audit and cross-judge are model-assisted (same vendor family), not human-adjudicated; the risk-stratified sample mixes quota-defined strata and clusters on 187 entities, and unparseable cross-judge responses may be non-random. (6) The isolated experiment uses a semantic-replica scanner (not instrumented against the service path) and committed-snapshot dictionaries over ten resamples of one corpus — a stability check; its metrics are structural, with semantic reading confined to sampled audits (the blind round included: same judge family, term-mention rubric not comparable to the link audit); governance verdicts (removed 0.033 vs retained 0.700) inherit that exposure. (7) The cross-domain study (§8) is model-annotated with a single-author human verification of a 197-term sample (precision bound; recall is referenced by a single-author 30-abstract from-scratch enumeration; §8): silver enumeration agreement across families is 0.41–0.55; an instrument-sensitivity rerun on the shared 30-document subset shows the arms' recall ordering stable under both references (F1 mid-band orderings in CS are instrument-conditional); one gateway judge is unusable for enumeration, and the LLM-direct arm's silver-P is same-family inflated — the cross-domain conclusions rest on structural metrics, recall orderings, and arbitrated blind validity, with the source-substitution adaptation (LLM knowledge seeds in place of seed transplant) disclosed. The recall funnel and the selection study (§8.1) inherit the silver instrument for their loss attribution and training labels; the selection arms' dev labels come from the cross-family judge, and the negative result is stated under those proxies.

REFERENCES

[1] K. Frantzi, S. Ananiadou, H. Mima. Automatic recognition of multi-word terms: the C-value/NC-value method. *Int. J. on Digital Libraries* 3(2), 2000, 115–130.

[2] Bai, J., Fan, W., Hu, Q., et al. AutoSchemaKG: Autonomous knowledge graph construction through dynamic schema induction from web-scale corpora. *Proc. 64th Annual Meeting of the ACL (ACL 2026)*, 2026, pp. 20557–20584. doi:10.18653/v1/2026.acl-long.942.

[3] DIAL-KG: schema-free incremental knowledge graph construction via dynamic schema induction and evolution-intent assessment. arXiv:2603.20059, 2026 (preprint).

[4] Large language models for knowledge graph extraction: a schema-constrained evaluation framework. *Proc. KG-LLM Workshop at LREC 2026*, 2026.

[5] Enhancing automatic term extraction with large language models via syntactic retrieval. *Findings of the ACL: ACL 2025*, 2025. doi:10.18653/v1/2025.findings-acl.516.

[6] D. Nadeau, S. Sekine. A survey of named entity recognition and classification. *Lingvisticae Investigationes* 30(1), 2007, 3–26.

[7] H. Paulheim. Knowledge graph refinement: A survey of approaches and evaluation methods. *Semantic Web* 8(3), 2017, 489–508.

[8] E. Breck, S. Cai, E. Nielsen, M. Salib, D. Sculley. Data validation for machine learning. *Proc. MLSys*, 2019.

[9] S. Schelter, D. Lange, P. Schmidt, M. Celikel, F. Biessmann. Automating large-scale data quality verification. *Proc. VLDB Endowment* 11(12), 2018, 1781–1794.

[10] A. Rigouts Terryn, V. Hoste, P. Drouin, E. Lefever. TermEval 2020: Shared task on automatic term extraction using the annotated corpora for term extraction research (ACTER) dataset. *Proc. 6th Int. Workshop on Computational Terminology (NLP-CT)*, 2020, 85–94.

[11] W. L. Hamilton, J. Leskovec, D. Jurafsky. Diachronic word embeddings reveal statistical laws of semantic change. *Proc. 54th Annual Meeting of the ACL*, 2016, 1489–1501.

[12] A. S. Lippolis et al. Ontology generation using large language models. arXiv:2503.05388, 2025.

[13] ISO 1087: Terminology work and terminology science — vocabulary. ISO, 2019.

[14] Anonymous. KnowledgeNexus: A traceable graph–vector knowledge infrastructure and adaptive multi-channel retrieval for nuclear science literature. Companion manuscript, under submission; preprint in the artifact release.